MIL-S-12991B(ME)
3 August 1971
SUPERSEDING
MIL-S-12991A(CE)
5 June 1959

MILITARY SPECIFICATION

SURVEYING SET, TOPOGRAPHIC COMPANY;

PACKAGING OF

1. SCOPE

1.1 This specification covers the requirements for the preservation, packaging, packing and marking of Topographic Company Surveying Set, SC 6675-97-CL-E27 (FSN 6675-641-3601).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-781	- Strapping, Steel, Flat and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Fiberboard.
PPP-C-1120	- Cushioning Material, Uncompressed Bound Fiber for Packaging.
PPP-P-40	- Packaging and Packing of Hand Tools.

Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.

FSC 6675

Source: https://assist.dla.mil -- Downloaded: 202640-17T23:55Z
Check the source to verify that this is the currentversion before use.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified to prove prior to starting production packaging and packing, that the applied preservation, packaging, packing and marking comply with the requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:55Z
Check the source to verify that this is the current version before use.

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3 Preservation and packaging. Preservation and packaging shall be level A or C as specified.

3.3.1 Level A.

3.3.1.1 Preservatives. Preservatives specified herein shall conform to the specifications listed in and shall be applied in accordance with MIL-P-116.

3.3.1.2 Methods of preservation. Methods of preservation specified herein by symbols shall conform to the applicable symbols listed in MIL-P-116.

3.3.1.3 Surveying instruments and accessories. Surveying instruments and accessories shall be preserved and packaged in accordance with MIL-S-3534, level A.

3.3.1.4 Axes, die sets, hammers, knives, machettes, marline spikes, oiler, screwdriver, machette sheathes, measuring tapes and vises. These items shall be preserved and packaged in accordance with PPP-P-40, level A.

3.3.1.5 Clamp handles, cloth, compass, file folders, drawing paper, drafting scale, stereoscopes and twine. The compass shall be placed in the case. The sheets of drawing paper of like description shall be wrapped with commercial kraft paper, 30 pound basic weight, to form a 48 sheet unit package. Each steroeoscope shall be placed in its carrying case. Each item, roll, box or package as applicable shall be individually preserved method IC-1.

3.3.1.6 Map cases, cord, crayons, envelopes, lead pencils, writing pads, mechanical pencils, protractors, plotting scales, thermometers and mailing tubes. Items of like description shall be preserved together method IC-1 in quantities specified for each set.

3.3.1.7 Paint and varnish brushes. The bristle end of each brush shall have a double thickness wrap of 30 pound basic weight commercial kraft paper. The wrap shall be folded back over the bristle and secured to the brush handle with tape.

3.3.1.8 Incandescent lamps. Lamps of like description shall be preserved together in a close-fitting box conforming to PPP-B-636, W6c, style optional. Blocking, cushioning or separators as applicable shall be provided to separate the lamps within the container. The box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:55Z
Check the source to verify that this is the current version before use.

3.3.1.9 Plywood. The plywood sheets shall be strapped together in quantities specified for each set. Edge protectors shall be provided to prevent damage to the edges. Strapping shall conform to QQ-S-781, type I, class B, .023 by 1/2-inch wide.

3.3.1.10 Repair kit. The metal components of the repair kit shall be coated with type P-7 preservative, wrapped with barrier material and inserted in the repair kit case. The case shall then be preserved method IC-1.

3.3.1.11 Battery power supply, folding chairs, signal lights, taping stool, tension handles and umbrellas. Chairs shall be folded to the minimum dimensions. The signal light, with lamps, shall be placed in the carrying case. The umbrella shall be collapsed to the minimum dimension. Each item shall then be preserved method III in a close-fitting box conforming to PPP-B-636, W5c, style optional. After closure the box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

3.3.1.12 Dry batteries. Batteries of like description shall be preserved together method III in a close-fitting box conforming to PPP-B-636, W5c, style optional. After closure the box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

3.3.1.13 Target set. Components of the target set shall be placed in the equipment case. The case containing the components shall be preserved method IA-14. The case shall serve as the inner container and the outer shall be a box conforming to PPP-B-636, W5c, style optional. After closure the outer container shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

3.3.1.14 Chronometer and chronometer case. The gimbaled suspension of the case, containing the chronometer, shall be rigidly locked in the transporting position. The cover shall be closed and secured with the keeper. Each case, containing a chronometer, prepared as specified shall be packaged in a box conforming to PPP-B-636, W5c, style optional. The box shall be lined with 4-inch thick bound fiber cushioning material conforming to PPP-C-1120, type II. The cushioning shall be applied to provide 4-inch cushioning protection for sides, top and bottom of the chronometer case. After closure the box shall be waterproof sealed with tape as specified for slotted style boxes in the appendix to the box specification.

Source: https://assist.dla.mil -- Downloaded: 20260-17T23:55Z
Check the source to verify that this is the current version before use.

3.3.1.15 Consolidated packaging. The coils of wire and plywood sheets shall not require consolidated packaging. All other components comprising a complete set shall be consolidated packaged in close-fitting boxes conforming to PPP-B-601, domestic type, style optional, of shape, size and packaging arrangement best suited to comply with the packing requirements specified herein. Strapping of boxes shall not be required.

3.3.2 Level C. The components of each complete surveying set shall be preserved and packaged to afford adequate protection against deterioration and damage during shipment from the supplier to the initial destination for immediate use.

3.4 Packing. Packing shall be level A or C as specified.

3.4.1 Level A. Components of each complete surveying set, preserved and packaged as specified in 3.3, shall be packed in a crate conforming to MIL-C-104, type I, class 1 or 2, style A. Blocking, bracing and anchoring of the contents within the crate shall be in accordance with the crate specification and MIL-STD-1186. Strapping shall be zinc coated.

3.4.2 Level C. Components of each complete surveying set, preserved and packaged as specified in 3.3, shall be packed or prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier and as a minimum, shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

3.5 Marking. Marking shall be in accordance with MIL-STD-129.

3.6 Workmanship. Workmanship shall provide maximum protection to the set components by application of the requirements specified herein to prevent corrosion, deterioration and mechanical and physical damage during handling, shipment and storage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:55Z
Check the source to verify that this is the current version before use.

4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects specified in 4.4.4. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Inspection stages. Inspection shall be in two stages as follows:

(a) The first stage shall include inspection of materials, methods, containers and markings inside the shipping container.
(b) The second stage shall include inspection of materials and markings after closing the container.

4.4.2 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level in table I. Acceptable quality level (AQL) shall be 2.5 percent defective.

Table I. Classification of Defects

Defect	Level A	Level C
101. Materials, methods or containers not as specified. Each incorrect material, method, or container shall constitute one defect.	X	
102. Surveying instruments not preserved and packaged as specified (3.3.1.3).	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:55Z
Check the source to verify that this is the current version before use.

Table I. Classification of Defects (continued)

Defect	Level A	Level C
103. Hand tools not preserved as specified (3.3.1.4).	X	
104. Compass and stereoscopes not placed in the cases and drawing paper not unit packaged as specified (3.3.1.5).	X	
105. Paint and varnish brushes not packaged as specified (3.3.1.7).	X	
106. Lamps not packaged as specified or box not waterproof sealed as specified (3.3.1.8).	X	
107. Edge protectors not provided for strapping of plywood or strapping not zinc coated as specified (3.3.1.9).	X	
108. Repair kit components not coated with preservative as specified (3.3.1.10).	X	
109. Chairs or umbrella not folded or collapsed, signal light not placed in case or boxes not waterproof sealed as specified (3.3.1.11).	X	
110. Box containing the dry batteries not waterproof sealed as specified (3.3.1.12).	X	
111. Outlet container for the IA-14 not waterproof sealed as specified (3.3.1.13).	X	
112. Gimbaled suspension of cases containing a chronometer not locked; 4 inch cushioning not provided and applied as specified and box not waterproof sealed as specified (3.3.1.14).	X	
113. Consolidated packaging not as specified (3.3.1.15).	X	
114. Blocking, bracing and anchoring not as specified or strapping not zinc coated (3.4.1).	X	
115. Marking illegible, incomplete, incorrect or missing.	X	X

5. PREPARATION FOR DELIVERY

(Not applicable to this specification.)

6. NOTES

6.1 Intended use. This specification is intended to be used for direct reference in contracts or orders for the preservation, packaging, packing and marking of Surveying Set, SC 6675-97-CL-E27 (FSN6675-641-3601).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:55Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Time frame required for submission of the preproduction pack (see 3.1).
(c) Level of preservation and packaging and level of packing required (see 3.3 and 3.4).

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack and mark the surveying sets in accordance with this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 6675-A116

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:55Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-12991B(ME) Surveying Set, Topographic Company; Packaging Of

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:55Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. A. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
ATTN: AMSME-RZG
FORT BELVOIR, VIRGINIA 22060

OFFICIAL BUSINESS

POSTAGE AND FEES PAID

DEPARTMENT OF THE ARMY
U. S. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
ATTN: AMSME-RZG
FORT BELVOIR, VIRGINIA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:55Z
Check the source to verify that this is the current version before use.